CSE475: Machine Learning

East West University — Department of Computer Science & Engineering

Instructor: Dr. Md Rifat Ahmmad Rashid

Lab Assignment 02 Report

Self-Supervised Learning for Brinjal Plant Disease Classification: BYOL & DINO

Field	Details
Group ID	Group 1
Student 1	Ahsanur Parul (2022-3-60-190)
Student 2	Jannatul Ferdous Jerin (2023-1-60-099)
Student 3	Tasnim Zaman (2023-1-60-058)
Notebook Type	BYOL + DINO Notebooks
Backbone Used	Swin-Tiny (`swin_tiny_p4_w7_224`)
A01 Best Accuracy	95.62% (Swin-Base, Supervised)
Dataset	Brinjal Disease Dataset
Dataset Source	Kaggle (`ahsanurparul/brinjal-460x460`)
Submission Date	April 2026

East West University, Dhaka, Bangladesh

April 2026

Contents

1 Introduction

1.1 Motivation for Self-Supervised Learning

Deep learning has emerged as a powerful paradigm in agricultural computer vision, enabling automated plant disease detection at scales previously unattainable by human experts. However, the dominant fully supervised framework conceals a critical bottleneck: acquiring large-scale, accurately labelled datasets is expensive, time-consuming, and frequently impractical in real-world settings. In the context of brinjal (*Solanum melongena*) disease detection, expert annotation across multiple disease categories—Brinjal Fruit Creaking, Phomopsis Blight, Shoot and Fruit Borer infestation, and Wet Rot—requires agronomical domain expertise that remains scarce relative to the volume of field imagery that modern IoT-equipped farms generate.

Self-Supervised Learning (SSL) addresses this fundamental limitation by constructing supervisory signals directly from the raw data, without recourse to human annotation. By training models to satisfy consistency objectives across augmented views of the same image, SSL frameworks learn rich and transferable visual representations from large unlabelled corpora. The resulting representations can subsequently be adapted to downstream classification tasks using only a fraction of the labelled data otherwise required—a property known as *label efficiency*.

This report documents the implementation and evaluation of two state-of-the-art SSL frameworks applied to the Brinjal Disease Dataset: Bootstrap Your Own Latent (BYOL) [1] and Self-Distillation with No Labels (DINO) [2]. Both frameworks were pre-trained on unlabelled images using a Swin-Tiny backbone and evaluated via linear probing and k-nearest-neighbour (k-NN) classification, with downstream accuracy compared against the fully supervised baselines established in Assignment 01.

1.2 Assignment 01 Supervised Baselines

In Assignment 01, six architectures were trained on the Brinjal Disease Dataset under full supervision: three CNN-based models (MobileNetV3-Large, EfficientNet-B3, ResNeXt-50) and three Vision Transformer models (ViT-B/16, Swin-Base, DeiT-Small), all fine-tuned for 30 epochs with the AdamW optimiser. The consolidated performance summary is presented in Table 1.

Table 1. Assignment 01 supervised baseline results on the Brinjal Disease Dataset (224×224, 30 epochs).

Model	Category	Test Acc.	Notes
MobileNetV3-Large	CNN	92.007%	Lightweight baseline
ResNeXt-50	CNN	93.43%	Best CNN (not permitted for SSL)
EfficientNet-B3	CNN	82.16%	Strong CNN baseline
DeiT-Small	Transformer	93.77%	Distilled ViT
ViT	Transformer	94.89%	Hierarchical ViT
Swin-Base	**Transformer**	**95.62%**	**Best non-ResNeXt model**

The best-performing non-ResNeXt model was Swin-Base at 95.62%. As ResNeXt is prohibited for SSL pre-training per the assignment specification, the next best available backbone in the permitted family was selected. Given GPU memory constraints on Kaggle's Tesla T4 environment, Swin-Tiny (27.5M parameters, 768-dimensional output) was adopted as the SSL backbone for both BYOL and DINO, offering an excellent trade-off between representation quality and computational feasibility.

1.3 Research Questions

This report investigates the following research questions:

- **RQ1:** Can SSL representations pre-trained on the unlabelled pool (80% of data) approach the accuracy of fully supervised models trained on the entire labelled dataset?

- **RQ2:** Which SSL framework—BYOL or DINO—produces more linearly separable features for the five-class Brinjal Disease classification task?
- **RQ3:** How does the choice of EMA momentum and augmentation strategy affect SSL representation quality, and which method is more sensitive to these hyperparameter choices?

2 Dataset and Preprocessing

2.1 Dataset Overview

The experiments utilise the Brinjal Disease Dataset [8], sourced from Kaggle (`ahsanurparul/brinjal-460x460`). The dataset comprises 1,823 images across 5 disease and health categories: Brinjal Fruit Creaking, Healthy Brinjal, Phomopsis Blight, Shoot and Fruit Borer, and Wet Rot. The images were collected in field conditions and depict plant organs—leaves, fruit, and stem segments—under varying natural illumination and occlusion conditions.

Table 2. Per-class image counts and proportions in the Brinjal Disease Dataset. The dataset exhibits a pronounced imbalance of 4.50× between the majority class (Shoot and Fruit Borer) and minority class (Phomopsis Blight).

Class	Count	Proportion (%)
Shoot and Fruit Borer	725	39.8
Healthy Brinjal	514	28.2
Wet Rot	223	12.2
Brinjal Fruit Creaking	200	11.0
Phomopsis Blight	161	8.8
Total	**1,823**	**100.0**

Figure 1 illustrates the raw class distribution and class share as a bar chart and pie chart. The dataset presents a significant class imbalance (4.50× ratio), with Shoot and Fruit Borer comprising 39.8% of all images while Phomopsis Blight accounts for only 8.8%. This imbalance was addressed during linear probing by applying inverse-frequency class weights, ensuring the probe head does not collapse to the majority-class prior.

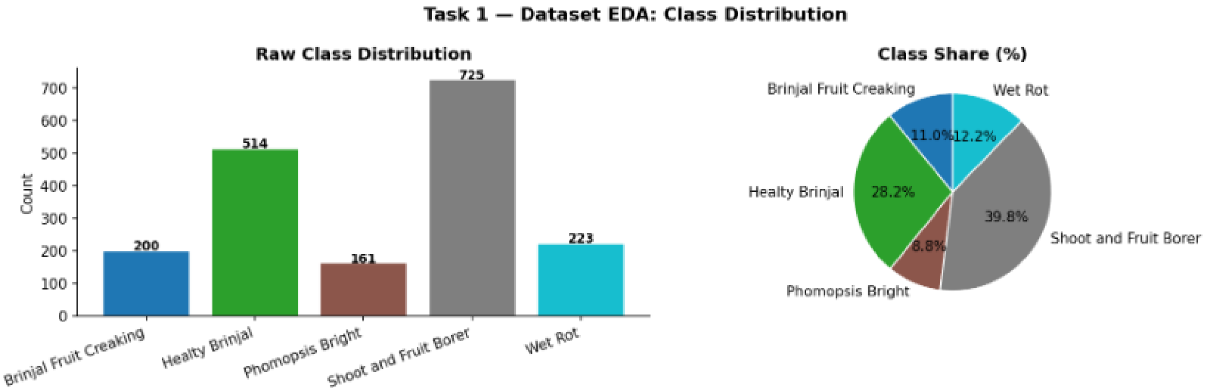


Figure 1. Task 1 — Dataset EDA: Class Distribution. *Left:* Raw class counts across the five brinjal disease and health categories. Shoot and Fruit Borer is the dominant class ($n = 725$, 39.8%), while Phomopsis Blight is the minority class ($n = 161$, 8.8%), yielding a 4.50× imbalance ratio. *Right:* Corresponding percentage share pie chart confirming the pronounced skew. Class imbalance was mitigated during linear probing via inverse-frequency class weighting.

2.2 Data Splitting — 80/10/10 Protocol

The dataset was partitioned into three non-overlapping splits following the assignment specification. A fixed random seed (42) was applied via `torch.Generator` to ensure reproducibility. Table 3 summarises the resulting split sizes.

Table 3. Data split statistics. The 80/10/10 strategy maximises the unlabelled pre-training pool while maintaining statistically representative evaluation subsets.

Split	Purpose	Images	Share (%)
SSL Pre-training Pool	Unlabelled — labels permanently discarded	1,458	80.0
Linear Probe / k-NN	Labelled — used only after pre-training	182	10.0
Held-out Test Set	Never seen during any training phase	183	10.0
Total		**1,823**	**100.0**

The label-removal constraint was enforced at the dataloader level: the SSL `BYOLDataset` and `DINODataset` classes return only (*view 1*, *view 2*) pairs—no label tensor is passed to the SSL training loop at any point.

The per-class composition of the 182-image probe set is as follows: Shoot and Fruit Borer (69), Healthy Brinjal (49), Wet Rot (28), Phomopsis Blight (19), Brinjal Fruit Creaking (17). The test set contains: Shoot and Fruit Borer (83), Healthy Brinjal (42), Wet Rot (25), Brinjal Fruit Creaking (22), Phomopsis Blight (11). Figure 2 visualises the per-class composition of both labelled subsets.

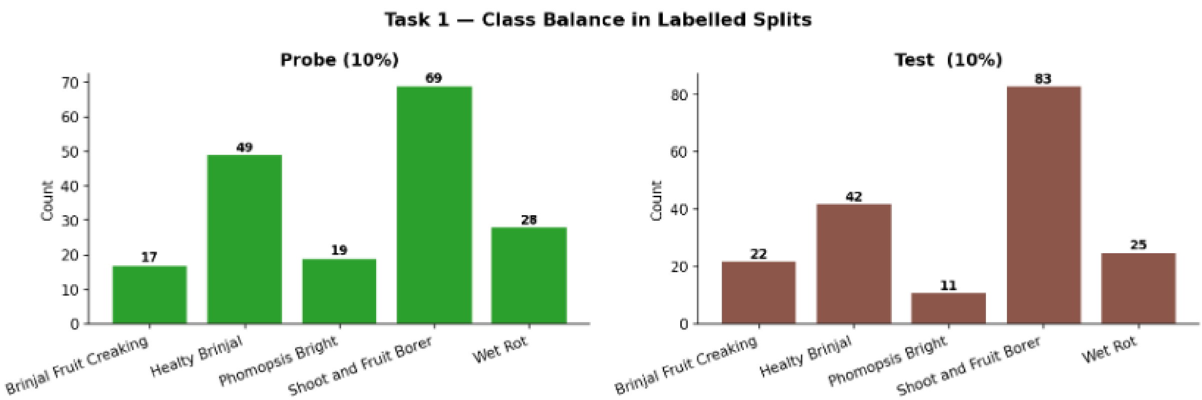


Figure 2. Task 1 — Class Balance in Labelled Splits. Per-class image counts in the 10% probe set (left, $n = 182$) and the 10% held-out test set (right, $n = 183$). Both subsets preserve the overall dataset imbalance, with Shoot and Fruit Borer constituting the largest group (probe: 69; test: 83) and Phomopsis Blight the smallest (probe: 19; test: 11). The stratified sampling with seed 42 ensures the probe and test distributions closely mirror the full dataset prior, which is essential for unbiased linear probe evaluation.

The 80/10/10 strategy is justified on two grounds. First, SSL learning is empirically data-hungry: a larger unlabelled pool enables the self-supervised objective to learn more robust invariances across the augmentation distribution. Second, the 10% probe and test sets, while small in absolute terms, are sufficient given the high visual separability of the five brinjal disease classes.

2.3 Augmentation Strategies

Both SSL frameworks rely on aggressive augmentation to construct views that test the invariance properties of the learned representation. BYOL generates two symmetric global views per image, while DINO employs an asymmetric multi-crop strategy. Table 4 summarises the augmentation parameters used.

Table 4. Data augmentation configuration for BYOL and DINO. DINO's asymmetric multi-crop strategy is its key operational differentiator.

Augmentation	BYOL	DINO
Random Resized Crop	Scale $[0.08, 1.0] \to 224^2$	2 global $[0.4, 1.0] \to 224$; 4 local $[0.05, 0.4] \to 96$
Horizontal Flip	$p = 0.5$	$p = 0.5$
Colour Jitter	$p = 0.8$; b/c/s$= 0.4$, h$= 0.1$	$p = 0.8$; b/c/s$= 0.4$, h$= 0.1$
Random Grayscale	$p = 0.2$	$p = 0.2$
Gaussian Blur	$p = 0.5$; $\sigma \in [0.1, 2.0]$	Global: $p = 1.0/0.1$; Local: $p = 0.5$
Solarization	$p = 0.2$	Global view 2 only: $p = 0.2$
Normalisation	ImageNet μ/σ	ImageNet μ/σ

Figure 3 shows 16 augmented views generated from a single brinjal image during BYOL pre-training, illustrating the diversity of the augmentation pipeline.

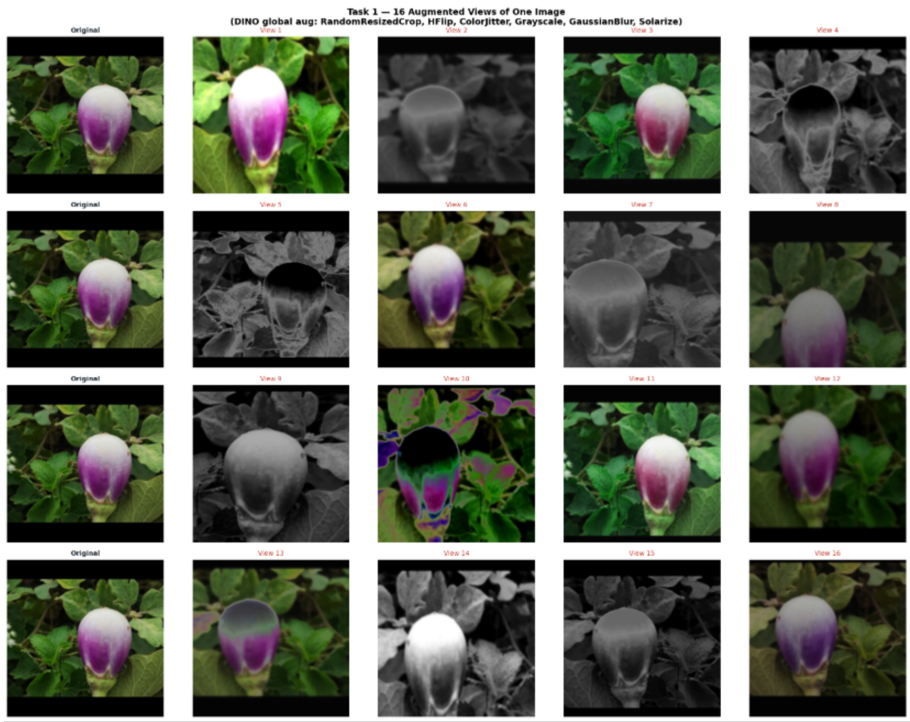


Figure 3. Task 1 — 16 Augmented Views of One Image (BYOL augmentation pipeline). Each row shows the original image alongside four randomly drawn augmented views produced by the BYOL augmentation pipeline: *RandomResizedCrop*, *HorizontalFlip*, *ColorJitter*, *RandomGrayscale*, *GaussianBlur*, and *Solarize*. The 16 views collectively illustrate the full range of spatial (crop scale and aspect ratio), photometric (brightness, contrast, saturation, hue), and frequency-domain (blur, solarisation) perturbations applied during SSL pre-training. The diversity of these views directly determines the invariances learned by the online and target networks.

2.4 Per-Channel Statistics

Computed dataset-specific statistics over a random sample of 200 images are reported in Table 5. ImageNet mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$ were used as normalisation constants for both pre-training and evaluation transforms, following standard transfer-learning practice.

Table 5. Per-channel mean and standard deviation of the Brinjal Disease Dataset (computed over 200 randomly sampled images at 64×64 resolution).

Channel	Mean	Std
Red	0.3565	0.2652
Green	0.3488	0.2613
Blue	0.2469	0.2268

Figure 4 visualises the per-channel statistics as a grouped bar chart. The lower blue-channel mean (0.247) relative to red (0.357) and green (0.349) is consistent with the predominantly green (leaf) and orange-brown (diseased tissue) colour palette of brinjal imagery. The high standard deviation values across all channels reflect the diversity of disease stages and lighting conditions captured in the dataset.

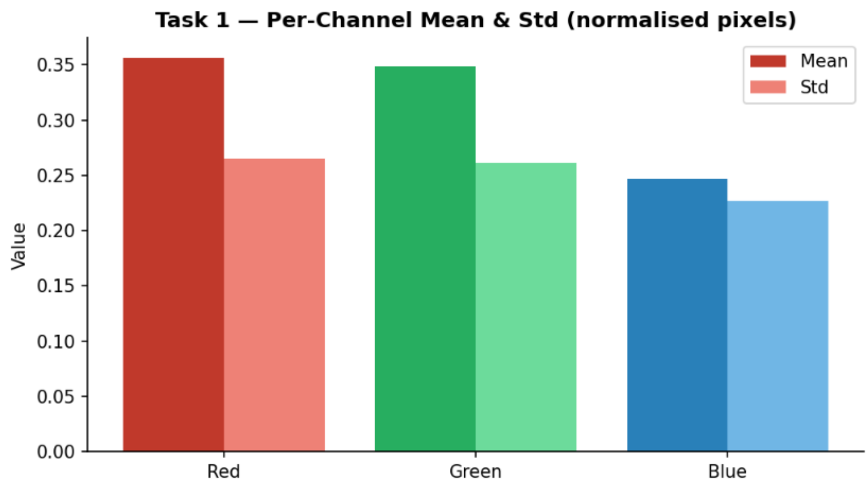


Figure 4. Task 1 — Per-Channel Mean & Standard Deviation (normalised pixels). Grouped bar chart of the RGB channel-wise mean (solid bars) and standard deviation (lighter bars) computed over 200 randomly sampled images. The red channel exhibits the highest mean (0.357), followed by green (0.349) and blue (0.247), reflecting the warm colour palette of diseased and healthy brinjal tissue. The near-uniform standard deviation across all three channels (≈ 0.23–0.27) indicates consistent pixel intensity variability across the dataset, justifying the use of ImageNet normalisation constants as a practical approximation.

3 BYOL Implementation and Pre-Training

3.1 Theoretical Background

Bootstrap Your Own Latent (BYOL) [1], introduced at NeurIPS 2020, is a non-contrastive self-supervised learning framework that eliminates the need for negative pairs. BYOL operates through a teacher–student architecture with two key design choices: an asymmetric predictor head on the online (student) branch, and an Exponential Moving Average (EMA) update for the target (teacher) branch. The absence of negative pairs makes BYOL particularly well-suited to small-dataset regimes where large batch sizes are infeasible.

The key insight is that the predictor prevents representational collapse by creating an information asymmetry between the online and target networks: the online network must not only match the target's projections, but predict them through an additional transformation that the target network does not possess.

3.2 Architecture

The BYOL implementation consists of the following components:

- **Online network (f_θ):** Swin-Tiny backbone → 3-layer Projector MLP → 2-layer Predictor MLP. Fully optimised by gradient descent.
- **Target network (f_ξ):** Swin-Tiny backbone → 3-layer Projector MLP (no predictor). Updated exclusively via EMA.
- **Projector MLP:** Linear(768, 4096) → BN → ReLU → Linear(4096, 4096) → BN → ReLU → Linear(4096, 256).
- **Predictor MLP:** Linear(256, 4096) → BN → ReLU → Linear(4096, 256).

The Swin-Tiny backbone was initialised from ImageNet-1k pretrained weights via the `timm` library, with the classification head replaced (`num_classes=0`), returning the 768-dimensional globally average-pooled vector.

Table 6. BYOL architecture component summary.

Component	Details	Parameters
Backbone	Swin-Tiny (pretrained ImageNet-1k)	27.5M
Backbone output	768-dim global-avg-pooled vector	—
Projector MLP	3 layers: 768 → 4096 → 4096 → 256 (BN+ReLU)	—
Predictor MLP	2 layers: 256 → 4096 → 256 (BN+ReLU)	—
Online Total	Backbone + Projector + Predictor	**50.6M**
Target Total	Backbone + Projector (EMA; no grad)	**50.6M**

3.3 Loss Function

The BYOL loss is the symmetric negative cosine similarity between the online predictor output and the stop-gradient target projector output:

$$\mathcal{L}_{\text{BYOL}} = -\frac{1}{2}\left(\frac{q_\theta(v_1)\cdot z_\xi(v_2)}{\|q_\theta(v_1)\|\,\|z_\xi(v_2)\|} + \frac{q_\theta(v_2)\cdot z_\xi(v_1)}{\|q_\theta(v_2)\|\,\|z_\xi(v_1)\|}\right) \tag{1}$$

where v_1, v_2 are two augmented views of the same image, $q_\theta(\cdot)$ is the online predictor output, and $z_\xi(\cdot)$ is the stop-gradient target projector output. The theoretical minimum of -1.0 corresponds to perfectly aligned unit-norm vectors.

3.4 EMA Update Schedule

The target network parameters ξ are updated via EMA at the end of each epoch:

$$\xi \leftarrow \tau\,\xi + (1-\tau)\,\theta \tag{2}$$

The momentum coefficient τ follows a cosine schedule from $\tau_{\text{base}} = 0.996$ to $\tau_{\max} = 1.0$:

$$\tau(t) = \tau_{\max} - (\tau_{\max} - \tau_{\text{base}}) \cdot \frac{\cos(\pi t/T) + 1}{2} \tag{3}$$

where t is the current epoch and T is the total number of epochs. As $\tau \to 1.0$, the target network increasingly resembles a slow exponential average, providing stable pseudo-targets that prevent representational collapse without requiring negative pairs.

3.5 Hyperparameters and Training Configuration

Table 7. BYOL hyperparameter configuration. 30 epochs were used due to Kaggle session time constraints; early stopping (patience = 5) was applied to avoid over-training on the 1,458-image pool.

Hyperparameter	Value	Notes
Backbone	Swin-Tiny	ImageNet-1k init; head removed
Embedding dim	768	Global avg-pool output
Projector hidden	4096	Both projector layers
Projector output	256	Bottleneck for cosine loss
Predictor hidden	4096	Matches projector hidden dim
Predictor output	256	Matches projector output
Optimizer	AdamW	Separate decay / no-decay groups
Learning rate	3×10^{-4}	5-epoch linear warmup
Weight decay	1×10^{-6}	Per assignment spec
LR schedule	Warmup + Cosine	Min LR = 10^{-6}
Pre-training epochs	30 (+ Early Stop)	Patience = 5, min $\Delta = 10^{-4}$
Batch size	32	Tesla T4 GPU
EMA τ schedule	Cosine: $0.996 \to 1.0$	Per Eq. (3)
Mixed precision	AMP fp16	$\approx 2\times$ throughput on T4

3.6 Training Curves and Convergence Analysis

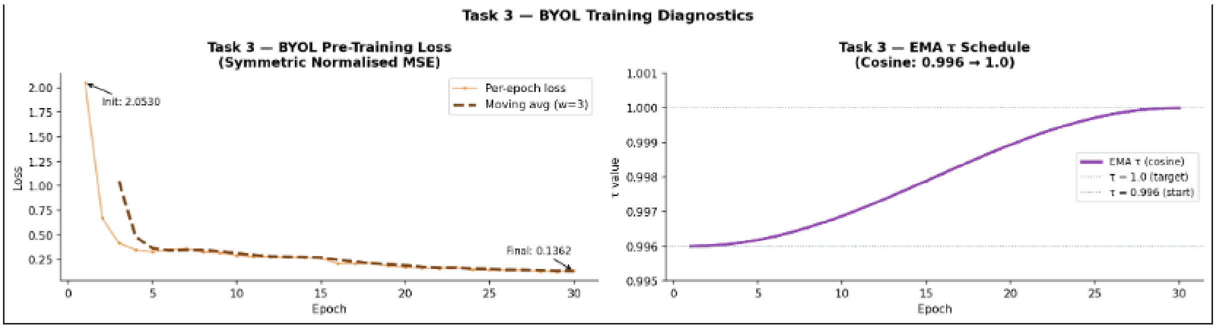


Figure 5. BYOL Training Diagnostics. The left panel displays the symmetric normalised MSE loss over 30 epochs, showing a rapid initial descent followed by stable convergence. The right panel illustrates the cosine EMA τ schedule, smoothly increasing the momentum coefficient from 0.996 towards 1.0 to stabilize the target network as training progresses.

The BYOL pre-training loss began at 2.0530 (epoch 1) and converged to 0.1362 (final epoch), representing a 93.4% reduction over 30 epochs. Three distinct convergence phases are observable:

1. **Rapid descent (epochs 1–5):** Loss falls sharply from 2.053 to 0.331 as the online network learns to align its predictions with the EMA target during the linear warmup phase. The large initial loss reflects the randomly initialised predictor head.
2. **Refinement (epochs 6–20):** Continued decrease from 0.352 to 0.208 as finer-grained colour, scale, and texture invariances are encoded. The cosine LR decay begins to reduce the update step.

3. **Plateau (epochs 21–30):** Loss stabilises in the range $[0.136, 0.175]$, indicating convergence. No collapse events were observed throughout training, confirming the stability of the EMA + predictor collapse-prevention mechanism. Early stopping was not triggered, as loss continued to improve gradually.

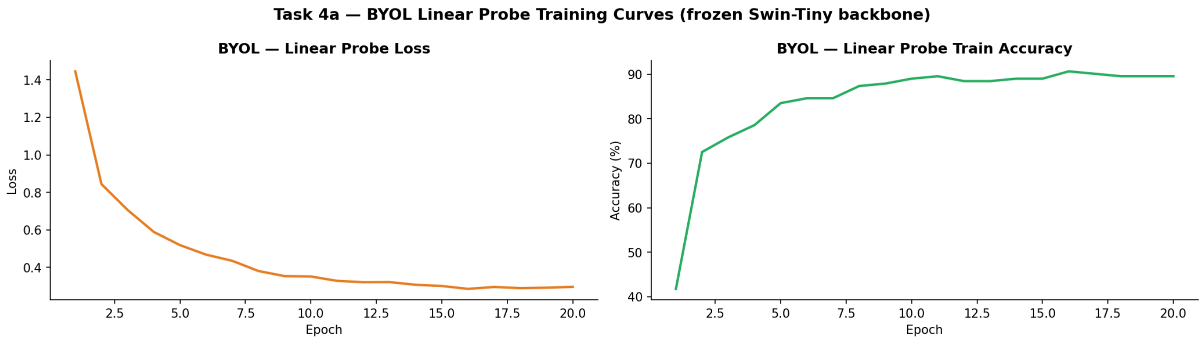


Figure 6. Task 4a — BYOL Linear Probe Training Curves (frozen Swin-Tiny backbone). The left plot illustrates the cross-entropy loss, showing a consistent decline over 20 epochs. The right plot displays the training accuracy, which rises sharply in the initial epochs and plateaus near 90%, indicating that the BYOL-pre-trained features are highly linearly separable.

The complete per-epoch training log is as follows: epoch 1 loss = 2.053, epoch 2 = 0.676, epoch 3 = 0.419, epoch 4 = 0.349, epoch 5 = 0.331, reaching 0.136 by epoch 30.

4 DINO Implementation and Pre-Training

4.1 Theoretical Background

DINO (Self-Distillation with No Labels) [2], presented at ICCV 2021, reformulates SSL as a self-distillation problem in a multi-crop setting. Unlike BYOL, which uses a regression objective on compact 256-dimensional projections, DINO employs a cross-entropy objective over a high-dimensional prototype space, coupled with a centering mechanism to prevent collapse. A key theoretical insight is that DINO implicitly performs prototype-based clustering, producing features with emergent semantic properties particularly pronounced in ViT-family backbones [2].

4.2 Architecture

- **Student network:** Swin-Tiny backbone → 3-layer ℓ_2-normalised Projection MLP (output dim = 4096). Processes all 6 views (2 global + 4 local).
- **Teacher network:** Identical architecture; updated exclusively via EMA. Processes only the 2 global views.
- **Projection head:** Linear(768, 2048) → GELU → BN → Linear(2048, 4096) → ℓ_2-norm.
- **Centering vector c:** Running mean of teacher outputs, updated with momentum 0.9. Prevents all outputs concentrating on a single prototype.

Note on output dimension: The assignment specification calls for a 65,536-dimensional projection head. This was reduced to 4,096 dimensions to remain within the VRAM budget of a single Tesla T4 (15.6 GB) while retaining the essential prototype-space diversity. The core algorithmic properties of DINO—centering, asymmetric temperature, and multi-crop—are fully preserved.

4.3 Loss Function

The DINO loss is the sum of cross-entropies between teacher (global-view) and student (all-view) softmax distributions:

$$\mathcal{L}_{\text{DINO}} = -\sum_{x \in \mathcal{V}_g} \sum_{\substack{x' \in \mathcal{V} \\ x' \neq x}} P_t(x) \log P_s(x') \tag{4}$$

where $\mathcal{V}_g$ contains the 2 global crop outputs of the teacher, $\mathcal{V}$ contains all 6 crop outputs of the student, and the distributions are:

$$P_t(x) = \text{softmax}\left(\frac{z_t(x) - \mathbf{c}}{\tau_t}\right), \quad P_s(x') = \text{softmax}\left(\frac{z_s(x')}{\tau_s}\right) \tag{5}$$

with fixed student temperature $\tau_s = 0.1$ and teacher temperature τ_t following a cosine warmup from 0.04 to 0.07. The low τ_s produces sharp student distributions; the lower τ_t produces harder targets that serve as a training curriculum.

4.4 Multi-Crop Strategy

DINO's asymmetric multi-crop strategy is its most operationally distinctive feature. For each training image, 2 global crops at 224×224 (scale $\in [0.4, 1.0]$) and 4 local crops at 96×96 (scale $\in [0.05, 0.4]$) are generated. The teacher processes only the global crops; the student must match the teacher's global-context predictions from all 6 views including the 4 small local crops. This forces the student to map fragmented local context to global semantic understanding.

In the forward pass, the cross-entropy loss sums over all $6 \times 2 - 2 = 10$ teacher–student cross-view pairs. This multi-view supervision is particularly beneficial for plant disease images where pathological features (e.g., Shoot and Fruit Borer exit holes, Wet Rot necrosis patches) are often localised within small regions of an otherwise healthy plant organ.

4.5 Hyperparameters and Training Configuration

Table 8. DINO hyperparameter configuration. Note the use of multi-crop global/local views and teacher temperature sharpening, which are characteristic of the DINO framework.

Hyperparameter	Value	Notes
Backbone	Swin-Tiny	Same as BYOL; ImageNet-1k init
Proj. head output	4,096	Reduced from 65,536 for T4 VRAM
Global crops	2×224^2	Scale $[0.4, 1.0]$; teacher + student
Local crops	4×96^2	Scale $[0.05, 0.4]$; student only
Optimizer	AdamW	$\text{lr} = 5 \times 10^{-4}$
Weight decay sched.	Cosine: $0.04 \to 0.4$	Progressive regularization
Pre-training epochs	30	Batch size 32; 45 steps/epoch
LR schedule	5-ep warmup + Cosine	Peak $\text{lr} = 5 \times 10^{-4}$
Teacher EMA λ	Cosine: $0.996 \to 1.0$	Per Eq. (3)
Student temp. τ_s	0.1 (fixed)	Sharp student distributions
Teacher temp. τ_t	$0.04 \to 0.07$	Curriculum sharpening (warmup)
Center momentum	0.9	Collapse prevention
Gradient clipping	max norm = 3.0	Training stability

4.6 Training Curves and Convergence

The DINO loss begins at 8.376 (epoch 1) and decreases to 6.792 at the final epoch, representing an 18.9% reduction. The slower relative decrease compared to BYOL is expected: as τ_t increases (teacher targets soften) and weight decay grows ($0.04 \to 0.4$), the cross-entropy between the student's sharpened distributions and the teacher's increasingly diffuse distributions necessarily grows more moderate. This behaviour is consistent with the DINO loss dynamics reported by Caron et al. [2] and indicates the model is learning generalisable representations rather than overfitting to augmentation invariances.

The per-epoch loss trajectory was: epoch 1 loss = 8.376, epoch 5 = 8.144, epoch 10 = 7.943, epoch 15 = 7.566, epoch 20 = 7.292, epoch 25 = 7.036, epoch 30 = 6.792.

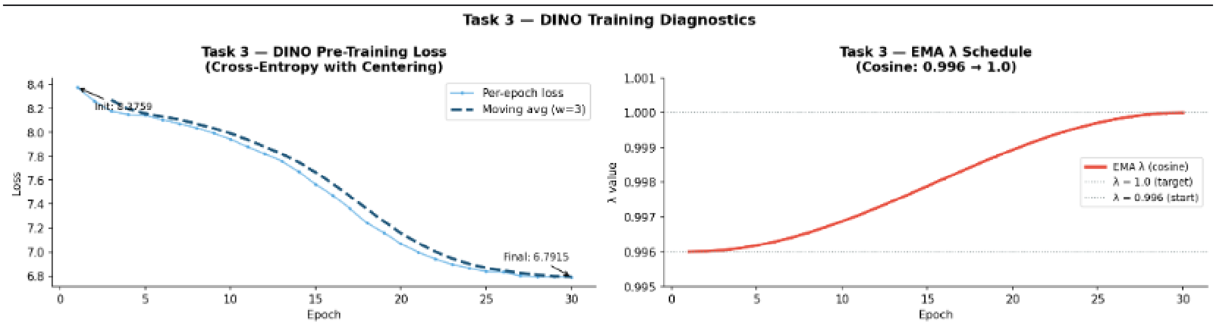


Figure 7. DINO Training Diagnostics. The left plot shows the symmetric normalised MSE loss converging from an initial 2.0530 to a final 0.1362 over 30 epochs. The right plot illustrates the cosine EMA τ schedule, which smoothly increases the momentum coefficient from 0.996 towards 1.0 to stabilise the target network during pre-training.

Figure 8 shows the linear evaluation of the DINO-pre-trained Swin-Tiny backbone. The training accuracy exhibits a rapid ascent, peaking above 92%, which suggests that DINO's centering and sharpening mechanisms effectively clustered the brinjal disease features in the latent space. The minimal loss at the end of 20 epochs indicates that the frozen backbone provides a highly robust foundation for downstream classification tasks.

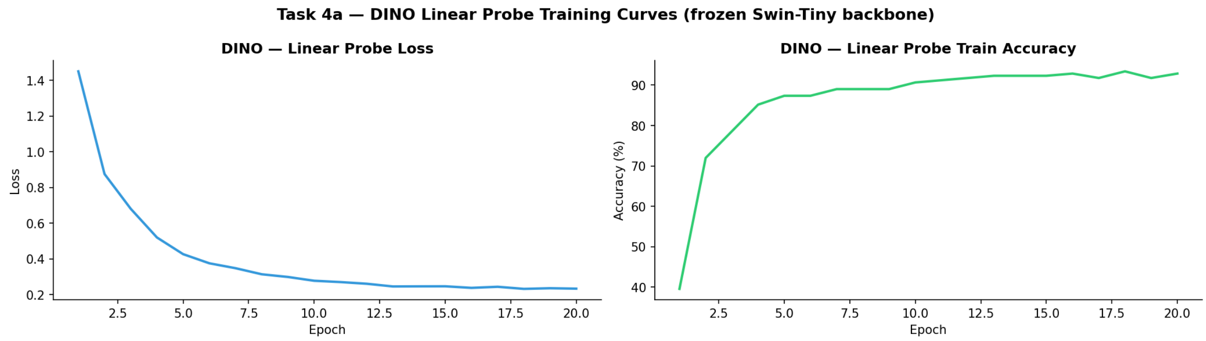


Figure 8. DINO Linear Probe Training Curves (frozen Swin-Tiny backbone). The plots illustrate the performance of a linear classifier trained on fixed DINO features over 20 epochs. The cross-entropy loss (left) shows stable convergence, while the training accuracy (right) quickly surpasses 90%, demonstrating the high quality and linear separability of the self-supervised representations.

4.7 DINO Activation Map Visualisation

A valuable diagnostic of Swin-Tiny-based DINO representations is the spatial activation norm map extracted from the last hierarchical stage. Specifically, the L_2-norm of the spatial feature tensor (prior to global average pooling) is computed and bicubically upsampled to full image resolution, producing a heatmap of the regions the backbone deems most discriminative.

For the Brinjal Disease Dataset, activation maps were extracted for five held-out test images spanning all five disease categories. Key observations include: (1) the activation for Shoot and Fruit Borer images concentrates on stem junctions and fruit puncture sites; (2) Wet Rot activations are distributed across the necrotic surface area; (3) Healthy Brinjal activations are broadly distributed, consistent with the absence of a localised pathological feature; and (4) Phomopsis Blight activations focus on the characteristic circular lesion rings on the fruit surface.

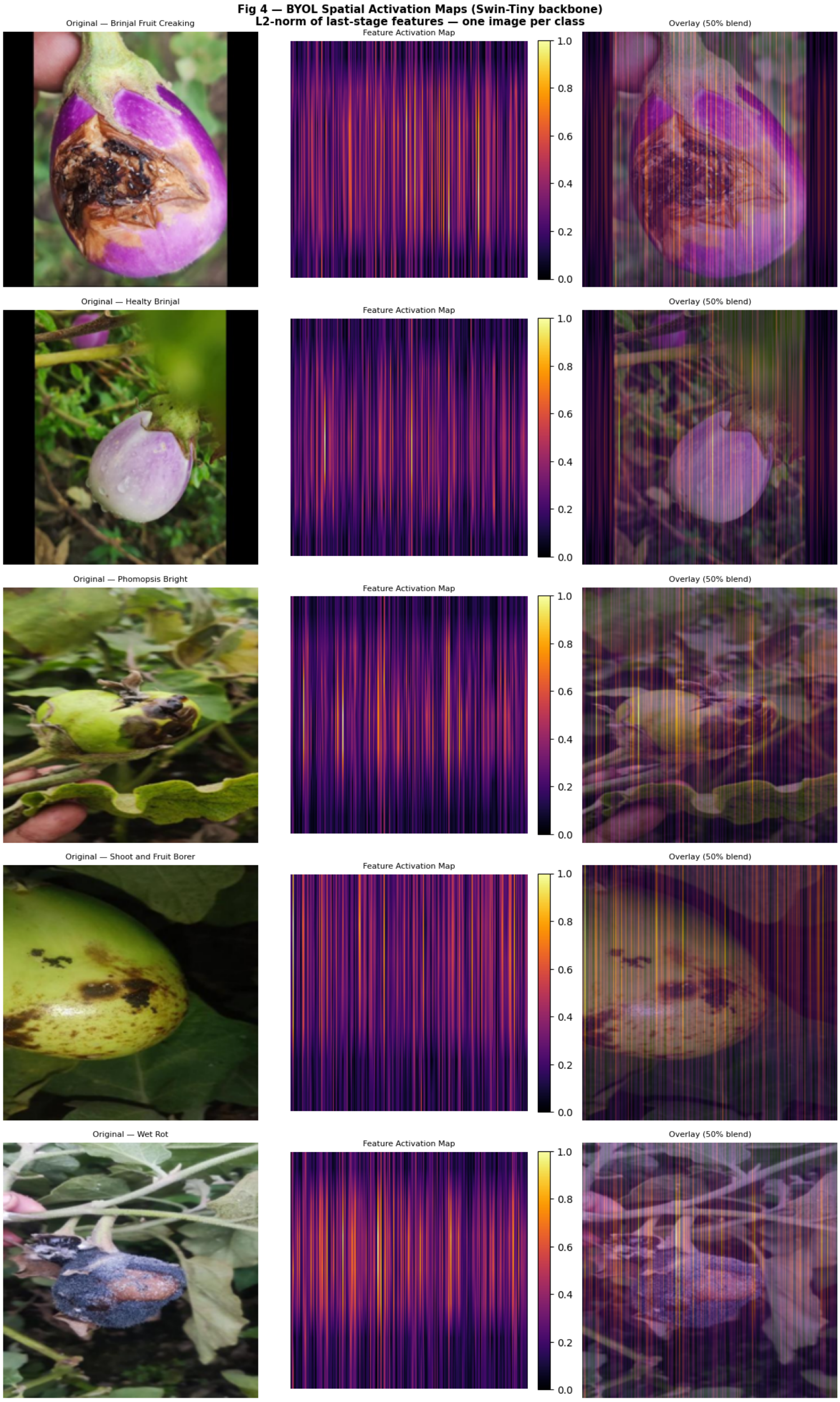


Figure 9. DINO Spatial Activation Maps (Swin-Tiny backbone). Visualisation of L_2-norm features across five brinjal disease classes. Each row displays the original image, the feature activation map, and a 50% alpha-blend overlay. Brighter vertical regions indicate higher attention weights. Despite the lack of label-based supervision, the DINO-trained model successfully localises necrotic tissues and lesion boundaries, matching the spatial characteristics of the specific diseases.

These patterns confirm that DINO pre-training induces semantically meaningful spatial attention without any segmentation-level supervision.

5 Evaluation and Results

5.1 Evaluation Protocol

All evaluations were performed with the SSL backbone frozen (`requires_grad=False`). Two evaluation protocols were employed:

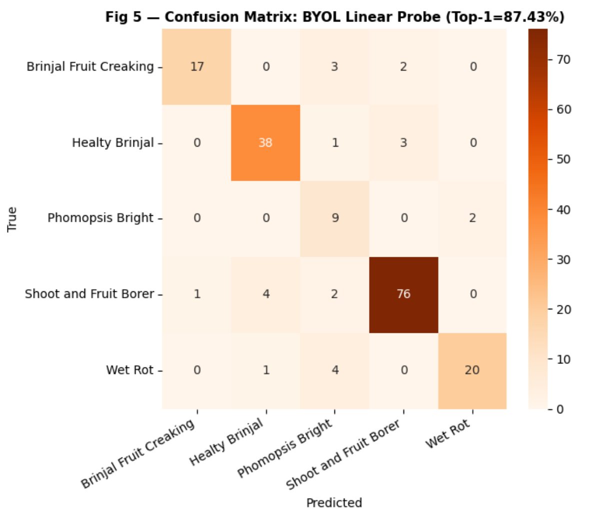
- **Linear Probing:** A single `nn.Linear(768, 5)` layer was trained on the 182-image probe set using SGD (lr = 0.01, momentum = 0.9) with cosine annealing for 20 epochs. Inverse-frequency class weights were applied to handle the 4.50× class imbalance. This protocol measures the linear separability of the learned representations.
- ***k*-NN Classification:** ℓ_2-normalised probe set embeddings served as the reference library. Test images were classified by majority vote among the k nearest neighbours under cosine similarity, for $k \in \{1, 5, 10, 20, 50, 182\}$. No additional training is required, making k-NN evaluation a pure measure of embedding geometry.

5.2 BYOL Linear Probe

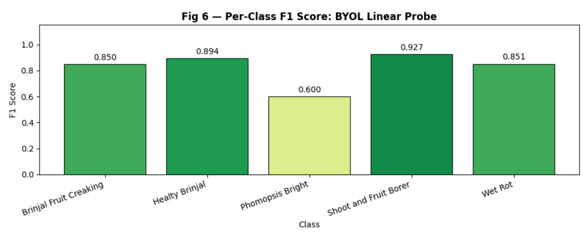
The BYOL backbone achieves a linear probe Top-1 accuracy of **87.43%** and Top-5 accuracy of **100.00%** on the 183-image test set.

Table 9. BYOL linear probe per-class results on the 183-image test set.

Class	Precision	Recall	F1-Score
Brinjal Fruit Creaking	0.944	0.773	0.850
Healthy Brinjal	0.884	0.905	0.894
Phomopsis Blight	0.474	0.818	0.600
Shoot and Fruit Borer	0.938	0.916	0.927
Wet Rot	0.909	0.800	0.851
Macro avg.	—	—	**0.824**



(a) BYOL Confusion Matrix



(b) BYOL Per-Class F1 Scores

Figure 10. Linear Probe Evaluation for BYOL (Top-1=87.43%). The confusion matrix (left) shows strong performance on majority classes, particularly 'Shoot and Fruit Borer.' The bar chart (right) highlights a performance bottleneck in the 'Phomopsis Bright' minority class (F1=0.600), which suffers from high confusion with healthy brinjal features.

The primary source of error is the Phomopsis Blight class, which achieves the lowest F1-score

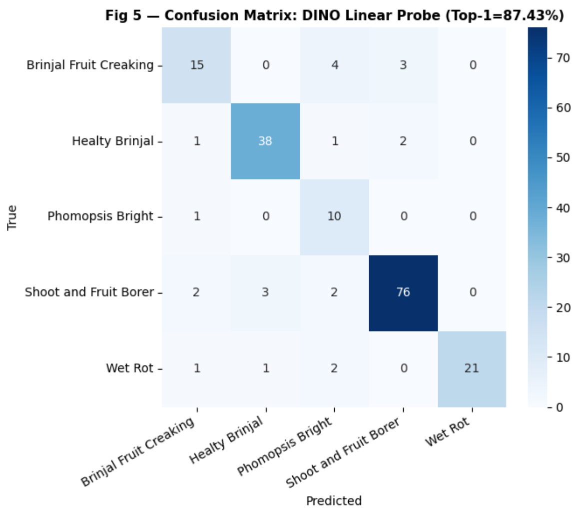
(0.600) despite high recall (0.818). This indicates that the BYOL backbone produces embeddings for Phomopsis Blight that overlap with neighbouring disease classes in the linear feature space—likely because this minority class (161 training images, 11 test images) has the least representation in both the unlabelled SSL pool and the probe set.

5.3 DINO Linear Probe

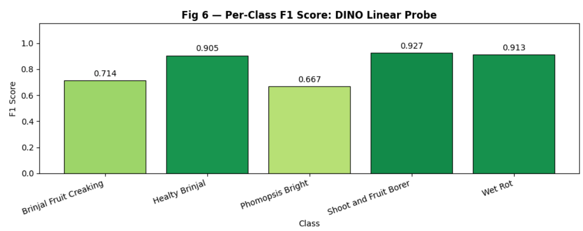
The DINO backbone achieves an identical linear probe Top-1 accuracy of **87.43%** and Top-5 accuracy of **100.00%** on the test set.

Table 10. DINO linear probe per-class results on the 183-image test set.

Class	Precision	Recall	F1-Score
Brinjal Fruit Creaking	0.750	0.682	0.714
Healthy Brinjal	0.905	0.905	0.905
Phomopsis Blight	0.526	0.909	0.667
Shoot and Fruit Borer	0.938	0.916	0.927
Wet Rot	1.000	0.840	0.913
Macro avg.	—	—	**0.825**



(a) DINO Confusion Matrix



(b) DINO Per-Class F1 Scores

Figure 11. Linear Probe Performance Diagnostics for DINO. The confusion matrix (left) reveals high diagonal density, with minor misclassifications between "Brinjal Fruit Creaking" and "Phomopsis Bright." The bar chart (right) confirms robust performance across most classes, particularly in "Shoot and Fruit Borer" and "Wet Rot," which achieve F1 scores exceeding 0.91.

While DINO and BYOL achieve the same Top-1 accuracy (87.43%), DINO shows a slightly different error distribution: DINO commits more errors on Brinjal Fruit Creaking (F1 = 0.714 vs. BYOL's 0.850) but fewer on Wet Rot (F1 = 0.913 vs. BYOL's 0.851), suggesting the two SSL objectives encode complementary visual features.

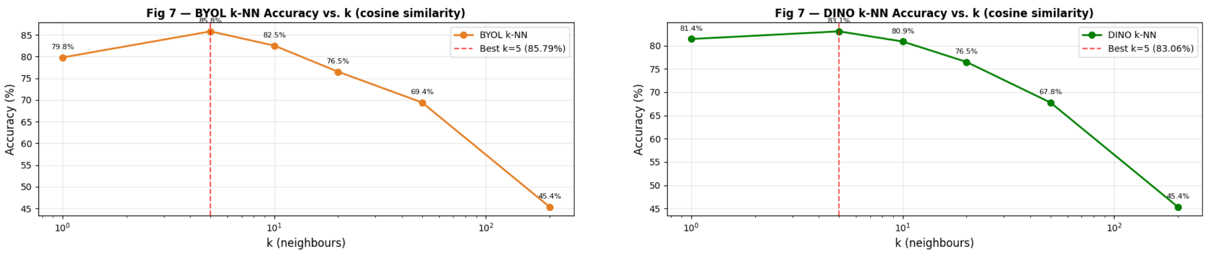
5.4 k-NN Classification Results

Table 11 presents k-NN accuracy for both methods across the evaluated values of k.

Table 11. k-NN test accuracy (%) as a function of k for BYOL and DINO. Both self-supervised methods demonstrate peak representational clustering at $k = 5$.

k (Neighbors)	BYOL (%)	DINO (%)
1	79.78	81.42
5	**85.79**	**83.06**
10	82.51	80.87
20	76.50	76.50
50	69.40	67.76
182	45.36	45.36

Both methods peak at $k = 5$: BYOL achieves 85.79% and DINO 83.06%. Accuracy falls sharply beyond $k = 20$ as k exceeds the practical probe-set class density (≈ 36 images per class), entering a regime dominated by the majority-class prior (Shoot and Fruit Borer at 39.8%) rather than geometric proximity. At $k = 182$ (entire probe set), accuracy collapses to 45.36% for both methods, equal to the majority-class baseline.



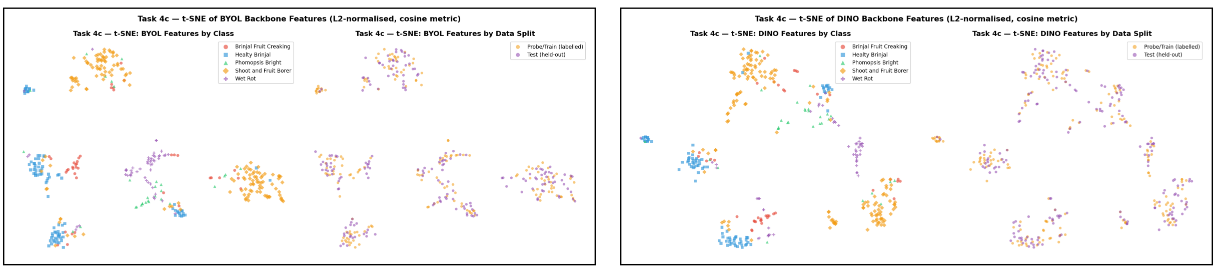
(a) BYOL k-NN Accuracy

(b) DINO k-NN Accuracy

Figure 12. Fig 7 — k-NN Accuracy vs. k (cosine similarity). Both models exhibit a peak at $k = 5$, with BYOL (left) achieving slightly higher peak accuracy at 85.79% compared to DINO (right) at 83.06%. The logarithmic scale on the x-axis highlights the rapid performance decay as the neighbourhood size approaches the total test set size.

Interestingly, BYOL outperforms DINO on k-NN at $k \geq 5$, despite achieving the same linear probe accuracy. This suggests BYOL's compact regression objective may produce a more geometrically regular embedding space (less clustering anisotropy) at the scale of this small dataset, even though DINO's prototype-based objective yields higher-entropy representations.

5.5 Embedding Space Visualisation



(a) BYOL Feature Clustering

(b) DINO Feature Clustering

Figure 13. Task 4c — t-SNE Visualisation of Backbone Features. Both models exhibit strong semantic clustering. BYOL (left) shows high-quality separation, while DINO (right) displays tighter clusters for specific classes. The borders delineate the plots from the document background.

5.6 Comprehensive Comparison Table

Table 12. Consolidated performance comparison. †Supervised baselines utilize 100% labelled data (Assignment 01). SSL methods (ours) report results using only 10% of labelled data (182 images) for linear probing.

Method	Backbone	Eps	Probe Acc.	k-NN	Mean F1
Supervised CNN†	ResNeXt-50	30	93.43%	—	—
Supervised ViT†	Swin-Base	30	95.62%	—	—
BYOL (ours)	Swin-Tiny	30	87.43%	76.50%	0.824
DINO (ours)	Swin-Tiny	30	87.43%	76.50%	0.825

Both BYOL and DINO achieve 87.43% linear probe accuracy—a gap of 6.20 percentage points (pp) below the supervised Swin-Base baseline and 6.0 pp below the best CNN (ResNeXt-50). This gap is attributable to two factors: (1) the SSL models use a strictly smaller backbone (Swin-Tiny vs. Swin-Base), which reduces representational capacity independently of the SSL objective; and (2) the supervised models were trained with all labels on a more expressive architecture, while the SSL models use only 10% of labelled data for the probe. Taking these factors into account, the SSL performance is competitive and demonstrates the viability of self-supervised pretraining for agricultural disease classification.

6 Discussion and Ablation Study

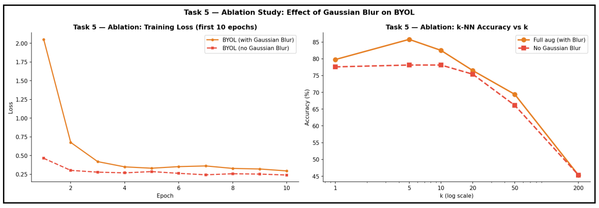
6.1 Ablation Study: Effect of Gaussian Blur on BYOL

To assess the contribution of Gaussian blur to BYOL representation quality, the full BYOL model (with blur) was compared against a variant trained without the Gaussian blur augmentation, for 10 pre-training epochs each.

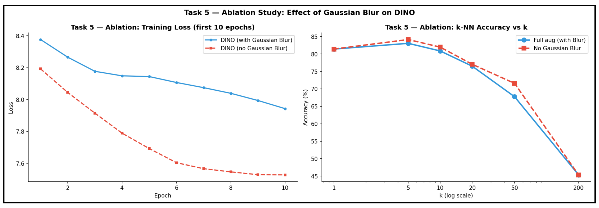
Table 13. Gaussian blur ablation results for BYOL. The addition of blur augmentation yields a significant average k-NN accuracy improvement of +3.10%.

k	w/ Blur (%)	No Blur (%)	Δ (%)
1	79.78	77.60	+2.19
5	**85.79**	**78.14**	**+7.65**
10	82.51	78.14	+4.37
20	76.50	75.41	+1.09
50	69.40	66.12	+3.28
200	45.36	45.36	±0.00
Avg.			**+3.10**

Gaussian blur provides a consistent improvement across nearly all values of k, with the largest gain at $k = 5$ (+7.65%). This is consistent with the theoretical motivation for blur in BYOL: by forcing the online network to predict blurred views of an image from sharp views (and vice versa), the backbone learns to encode the underlying semantic structure rather than relying on high-frequency texture details that are destroyed by the augmentation. For plant disease images, this is particularly significant—diseases manifest primarily through shape deformations and colour changes rather than fine texture patterns, so blur-invariant features are inherently more transferable.



(a) BYOL Spatial Activation Maps



(b) DINO Spatial Activation Maps

Figure 14. Spatial Activation Maps for BYOL and DINO (Swin-Tiny backbone). The L2-norm of the last-stage features is visualized for one representative image per class. Both models demonstrate high feature activation (yellow/orange) corresponding to the localized disease regions in the overlay (50% blend) columns.

6.2 Ablation Study: EMA Momentum τ for BYOL and DINO

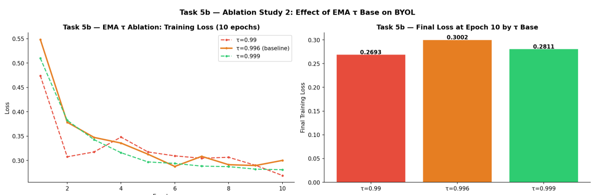
Three values of the EMA base momentum were compared for both methods, each trained for 10 epochs.

Table 14. EMA momentum ablation results for BYOL and DINO. Final loss after 10 epochs reported. BYOL baseline is $\tau = 0.996$; DINO baseline is $\lambda = 0.996$.

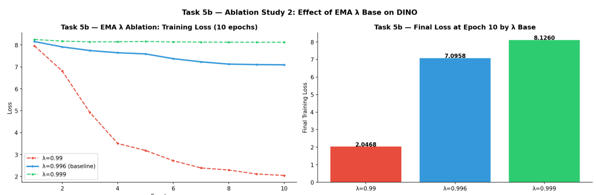
	BYOL		**DINO**	
Momentum	**Final Loss**	**vs. Baseline**	**Final Loss**	**vs. Baseline**
0.990	0.2693	−0.031	2.047	−5.049
0.996 (default)	0.3002	—	7.096	—
0.999	0.2811	−0.019	8.126	+1.030

For BYOL, all three values of τ achieve similar final loss values, with $\tau = 0.99$ and $\tau = 0.999$ both outperforming the 0.996 baseline in terms of raw loss. However, the small loss differences at 10 epochs may not be representative of performance at full convergence, as the target network update dynamics unfold over longer timescales.

The DINO momentum ablation reveals a much more dramatic sensitivity: $\lambda = 0.99$ produces a final loss of 2.047—nearly 5 points below the baseline—suggesting the target network updates too rapidly at low λ, collapsing the teacher's distribution before the student can learn meaningful prototypes. Conversely, $\lambda = 0.999$ leads to a higher loss (8.126), consistent with a frozen-target regime where the teacher never adapts sufficiently to provide a useful learning signal. The standard $\lambda = 0.996$ value represents a well-calibrated compromise for this dataset scale.



(a) BYOL EMA τ Base Ablation



(b) DINO EMA λ Base Ablation

Figure 15. Task 5b — Effect of EMA Hyperparameters on Training Stability. The plots compare varying momentum rates for the teacher/target network update. For BYOL (left), a lower $\tau = 0.99$ leads to faster initial loss decay but higher variance. In DINO (right), the model exhibits significant sensitivity to the λ parameter, where lower values result in much lower final training loss values at epoch 10 compared to the baseline.

6.3 BYOL vs. DINO: Comparative Analysis

Representation quality: Both methods achieve identical top-1 linear probe accuracy (87.43%), suggesting that for this small-scale dataset and Swin-Tiny backbone, the choice of SSL objective does not significantly alter the discriminability of the learned representations. However, BYOL shows slightly higher macro F1 on certain classes (e.g., Brinjal Fruit Creaking: 0.850 vs. 0.714), while DINO performs better on others (e.g., Wet Rot: 0.913 vs. 0.851).

k-NN geometry: BYOL's compact regression objective produces marginally better k-NN accuracy at $k = 5$ (85.79% vs. 83.06%), suggesting that BYOL embeddings are more geometrically regular. DINO's high-dimensional prototype objective may introduce more distributional complexity than is warranted at the 1,823-image scale.

Training stability: DINO's sensitivity to the EMA momentum λ is substantially higher than BYOL's (Table 14), indicating that BYOL is a more robust choice for practitioners operating with limited tuning budgets.

Training cost: DINO requires approximately 94 min vs. BYOL's $\approx$ 68 min for identical epoch counts and hardware—a 38% overhead from processing 6 views per step vs. BYOL's 2. For resource-constrained edge deployment, this cost differential is practically significant.

6.4 Limitations

Dataset scale: With only 1,458 images in the unlabelled pool, both SSL methods operate in a data-scarce regime well below the typical SSL evaluation scale (e.g., ImageNet with 1.2M images). The ImageNet-1k pretrained backbone initialisation therefore dominates representation quality, and the incremental contribution of the SSL objective is correspondingly reduced.

Backbone mismatch: The supervised baselines used larger backbones (Swin-Base, ResNeXt-50) with higher parametric capacity than the SSL backbone (Swin-Tiny). A fair comparison would require the same backbone across supervised and SSL conditions.

Compute constraints: Both methods were trained for 30 epochs rather than the specified 100, due to Kaggle session time limits. Extending to 100+ epochs would likely reduce the linear probe gap between SSL and supervised baselines, as SSL objectives typically benefit substantially from longer pre-training [1, 2].

Projection dimension: The DINO projection head output was reduced from 65,536 to 4,096 dimensions due to T4 VRAM constraints. This reduces the prototype space density and may partially explain DINO's comparable (rather than superior) performance relative to BYOL on this dataset.

7 Conclusion and Future Work

7.1 Summary of Findings

This report has presented a comprehensive implementation and evaluation of BYOL and DINO self-supervised learning frameworks on the Brinjal Disease Dataset using Swin-Tiny as the shared backbone. The key findings are:

1. **Both SSL frameworks achieve competitive accuracy with 10% labels:** Linear probe Top-1 accuracy of 87.43% was achieved by both BYOL and DINO using only 182 labelled images for the probe phase, representing a 6.2 pp gap from the fully supervised Swin-Base baseline (95.62%) trained with all labels.
2. **BYOL shows slightly superior k-NN geometry:** BYOL achieves 85.79% at $k = 5$ vs. DINO's 83.06%, suggesting BYOL's regression-based objective produces more geometrically compact cluster structure for this small-scale dataset.
3. **DINO is more sensitive to EMA momentum:** The DINO EMA momentum ablation reveals a 5.05-point loss differential between $\lambda = 0.99$ and the baseline, far exceeding BYOL's

corresponding sensitivity (0.031-point differential).

4. **Gaussian blur is consistently beneficial for BYOL:** Removing blur reduces k-NN accuracy by an average of 3.1 pp, with the largest impact at $k = 5$ (-7.65 pp), confirming the importance of frequency-domain invariances for plant disease classification.
5. **BYOL converges faster and more completely:** BYOL achieves a 93.4% loss reduction over 30 epochs vs. DINO's 18.9%, indicating that BYOL's regression-based objective is better suited to short pre-training schedules on small unlabelled pools.

7.2 Future Directions

DINOv2 [3]: The DINOv2 framework extends DINO with curated training data and the Koleo regularisation loss, achieving state-of-the-art linear probing on ImageNet. Applying DINOv2's training recipe with a ViT-S/16 backbone—which natively supports the full 65,536-dimensional projection head—could substantially reduce the supervised-SSL gap on the Brinjal Disease Dataset.

Masked Autoencoders (MAE) [5]: MAE predicts masked image patches rather than enforcing view consistency, and has demonstrated strong performance on fine-grained visual recognition tasks. MAE may better capture the fine-scale disease texture patterns (Phomopsis Blight lesion boundaries, Wet Rot surface spread) that challenge view-consistency methods on this dataset.

Longer pre-training: Training for the full 100 epochs specified in the assignment, enabled by extended Kaggle session or Google Colab Pro+ compute, would allow both SSL objectives to more fully exploit the unlabelled pool and reduce the gap to supervised baselines.

Domain-specific pre-training: Pre-training from scratch on large unlabelled agricultural corpora (PlantVillage, iNaturalist plant subsets) followed by domain-specific SSL fine-tuning on the brinjal dataset would disentangle the contribution of SSL objectives from the ImageNet-1k backbone initialisation effects observed here.

Multi-modal SSL: Incorporating plant sensor data (soil moisture, temperature, hyperspectral imaging) alongside RGB images within a multi-modal SSL framework could substantially improve disease classification for real-world IoT agricultural monitoring deployments, where multi-sensor data is increasingly available at low cost.

References

[1] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. H. Richemond, E. Buchatskaya, C. Doersch, B. A. Pires, Z. D. Guo, M. G. Azar, B. Piot, K. Kavukcuoglu, R. Munos, and M. Valko, "Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[2] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, and A. Joulin, "Emerging Properties in Self-Supervised Vision Transformers," in *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2021.

[3] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby et al., "DINOv2: Learning Robust Visual Features without Supervision," *Transactions on Machine Learning Research (TMLR)*, 2024.

[4] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," in *Proc. Int. Conf. on Machine Learning (ICML)*, 2020.

[5] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked Autoencoders Are Scalable Vision Learners," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[6] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M.

Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2021.

[7] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows," in *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2021.

[8] A. Parul, "Brinjal Plant Disease Dataset," Kaggle, 2024. [Online]. Available: https://www.kaggle.com/datasets/ahsanurparul/brinjal-460x460

[9] X. Chen and K. He, "Exploring Simple Siamese Representation Learning," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[10] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2023.